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Lab 6: MTS Tester and Digital Image Correlation

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Overview

In Lab 7, we use Digital Image Correlation (DIC) on the high-speed videos we captured during the Lab 6 tensile tests to see exactly how each part of the sample moves and stretches. First, we break a reference image into a grid of small patches and then, for each patch, find where it went in the next frame by matching the pixel patterns. Tracking those patch shifts frame by frame gives us a complete map of how the material displaces and stretches (for example, calculating ϵ_{11}) across the surface. This lets us spot exactly where the strain builds up, especially around notches, and then check if those hotspots line up with where the sample actually broke.

Experimental Setup

We first coated the ABS and TPU samples with a bold, spray-painted speckle pattern so the DIC software could track the surface. We then filmed each tensile test from start to finish with a high-resolution camera at a steady frame rate. After cutting each video to run from just before we applied load until just after the sample broke, we loaded them into MATLAB and ran Ncorr (an open-source DIC toolbox). In Ncorr, we set up a grid of small square patches (for example, 31×31 pixels with 5-pixel overlap) and used a matching method called ZNSSD to find how each patch moved between frames. From those movements, we built ϵ_{11} strain maps for three important moments in each test—early loading, halfway through, and just before failure—and then exported those maps for our plots and analysis.

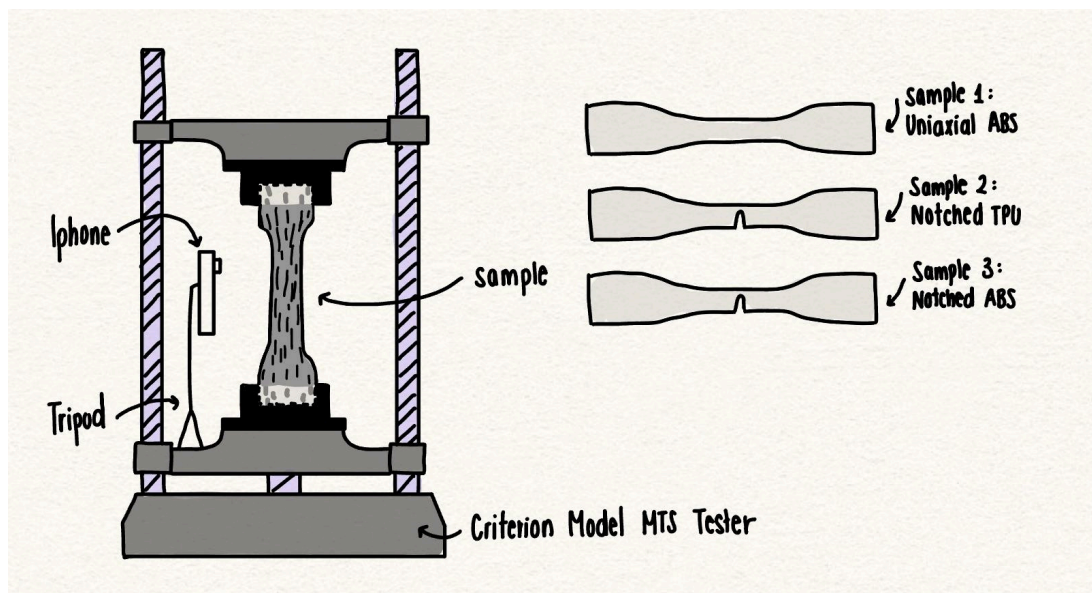


Figure 1 shows the experimental setup.

Strain the image along the Y-axis for ABS - Unnotched using Eulerian strain maps multiple times

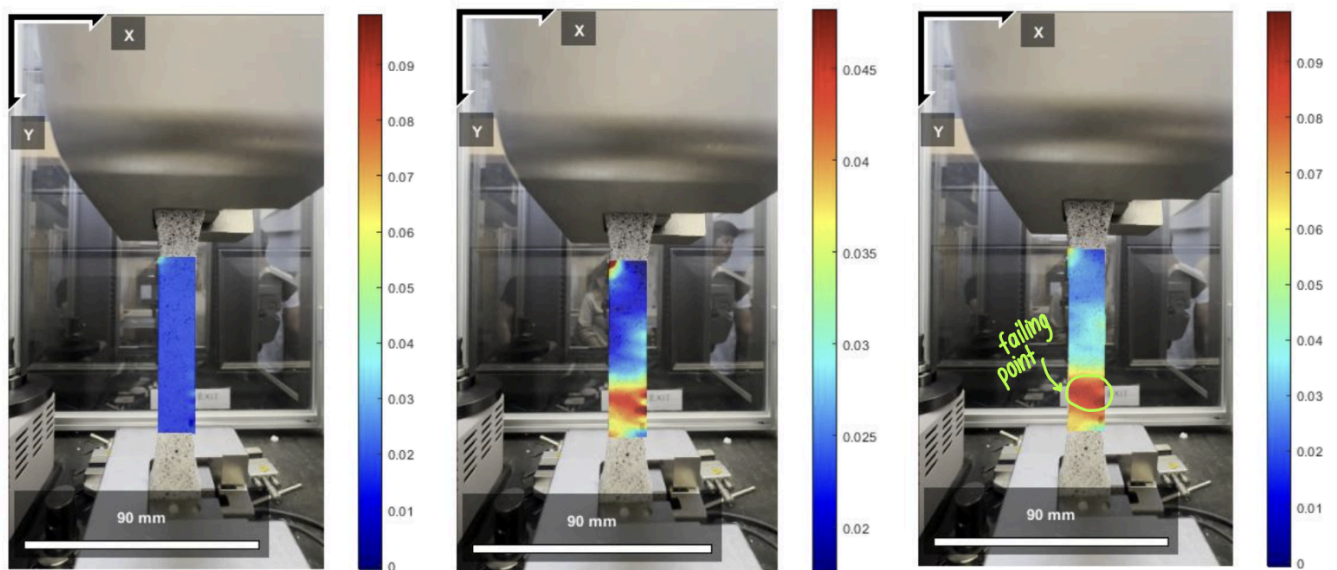


Figure 2. Panels (a–c) arranged left to right show properly scaled Eulerian ϵ_{11} strain maps of the **unnotched** ABS sample at three points in the tensile test, chosen to display notably different magnitudes:

- **(a) Left (Frame 28/50, $\approx 1/2$ through test), $\max \epsilon = 0.0471$**
 - Early strain concentration at the specimen ends
 - Moderate peak strain indicating the onset of yielding
- **(b) Middle (Frame 35/50, mid-test), $\max \epsilon = 0.0257$**
 - Noticeable drop in peak strain as deformation shifts toward the center
 - Strain is still highest at the ends but begins to spread inward
- **(c) Right (Frame 50/50, just before failure), $\max \epsilon = 0.0991$**
 - Intense, highly localized strain at the circled central region (actual crack-initiation site)
 - Peak strain more than doubles, signaling imminent rupture

From these maps, one would predict failure where ϵ_{11} concentrates most, in the central region, precisely where the sample fractures, confirming that DIC effectively forecasts the initiation site.

The strain map makes it look as though the ends of the unnotched ABS sample, right under the grips, are under the greatest deformation, so you'd naturally assume that's where the fracture would begin. In contrast, the actual break appeared near the mid-section of the bar, far from the obvious surface strain peaks.

This surprising result suggests that there are hidden influences beyond the surface strain contours. One possibility is that the 3D-printed ABS contains internal voids or density variations that weaken the center, even though its surface appears intact. Another is that the gripping jaws introduce uneven pressures or slight misalignments, creating complex subsurface stress regions that DIC doesn't capture. Lastly, because DIC only tracks what happens at the specimen's surface, any flaws or stress risers below the speckle pattern would go unnoticed, but could easily dictate where the crack starts.

Strain image along the Y-axis for TPU-Notched using Eulerian strain maps multiple times

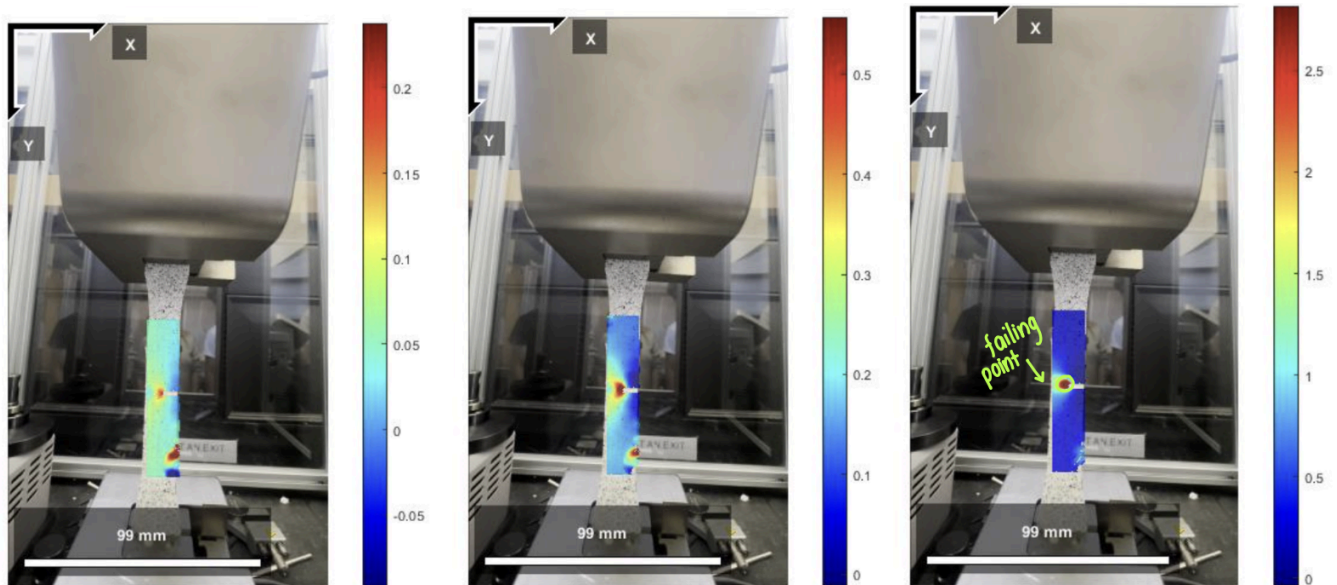


Figure 3. Panels (a–c) arranged left to right show properly scaled ϵ_{11} Eulerian strain maps of the notched TPU sample at three points in the tensile test, chosen to display notably different magnitudes:

- (a) Left (Frame 17/50, early test), max $\epsilon_{11} = 0.3429$
 - Initial strain concentration at the notch root in the elastic-to-plastic transition
 - Relatively modest peak strain compared to later stages
- (b) Middle (Frame 32/50, mid-test), max $\epsilon_{11} = 0.7263$
 - Pronounced plastic zone forming around the notch, with strain spreading along the gauge section
 - Peak strain more than doubles, signaling extensive yielding
- (c) Right (Frame 50/50, just before failure), max $\epsilon_{11} = 5.4649$
 - Extremely intense, highly localized strain at the circled crack-initiation site
 - Peak strain skyrockets, indicating imminent rupture

From these maps, one would predict failure at the notch root where ϵ_{11} concentrates most intensely, and this prediction matches the observed fracture location, confirming that full-field DIC strain mapping effectively forecasts the initiation site in TPU.

Strain image along the Y-axis for ABS - Notched using Eulerian strain maps multiple times.

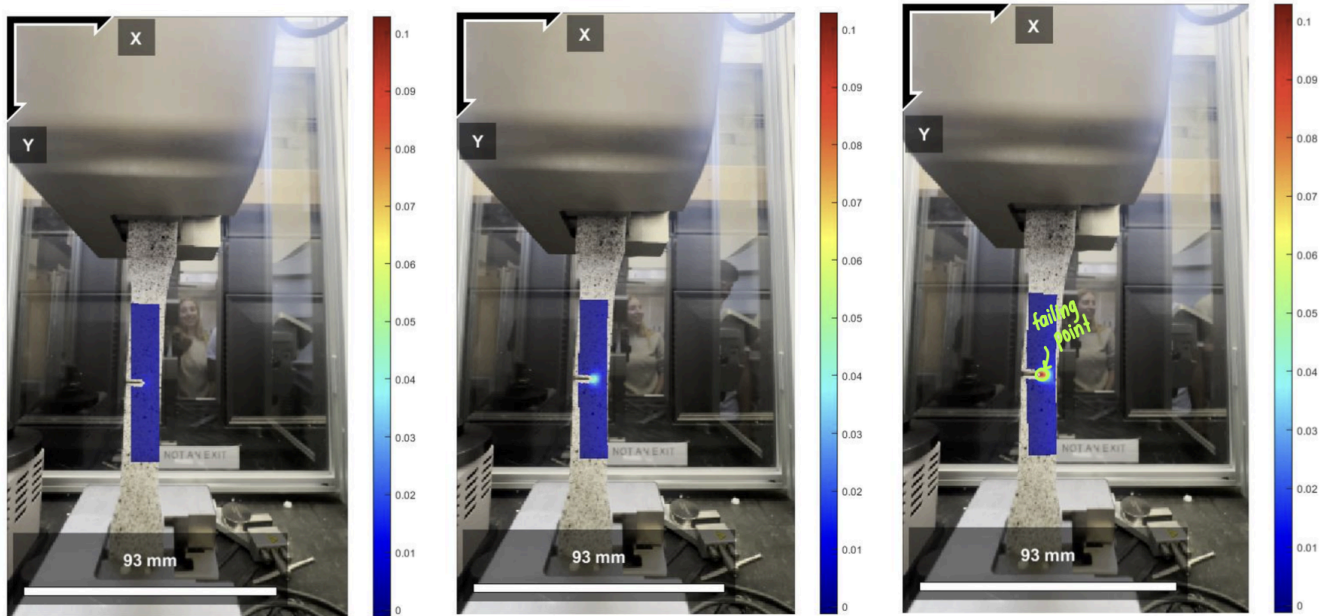


Figure 3. Images (a–c) arranged left to right show properly scaled Eulerian ϵ_{11} strain maps of the **notched** ABS sample at three points in the tensile test, chosen to display notably different magnitudes:

- **(a) Left (Frame 18/50, early test), $\max \epsilon_{11} = 0.0118$**
 - Very low strain localized at the notch root, indicating mostly elastic behavior
 - No significant plastic deformation yet
- **(b) Middle (Frame 36/50, mid-test), $\max \epsilon_{11} = 0.0499$**
 - Clear plastic zone forming around the notch with elevated strain levels
 - Strain begins to spread into the surrounding material
- **(c) Right (Frame 49/50, just before failure), $\max \epsilon_{11} = 0.1029$**
 - Very intense, highly localized strain at the circled crack-initiation site
 - Peak strain more than doubles from (b), signaling imminent fracture

By tracking the ϵ_{11} distributions from early yielding through to just before rupture, the maps consistently show the most intense strain localizing at the notch root. This progressive concentration—from initial elastic deformation, through the development of a semicircular plastic zone mid-test, to the sharply peaked strain at Frame 49/50—points unequivocally to the notch tip as the weakest point. When the samples finally fractured, the crack opened precisely at that same notch root hotspot, demonstrating that full-field DIC not only reveals where plasticity first accumulates but also reliably predicts the exact location of failure initiation.

Estimating and Reporting Strain Right at the TPU Notch, Right Before Failure

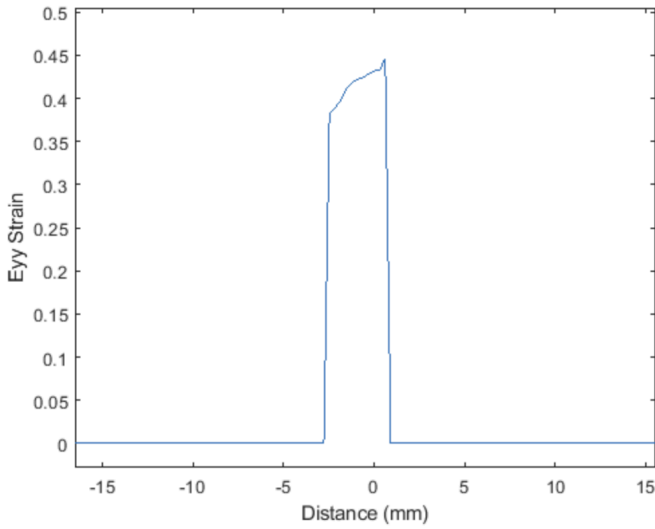


Figure 5. The ϵ_{yy} profile along a horizontal line just beside the notch, taken from the final pre-failure frame, shows an abrupt jump to about 0.083 at the notch location, climbing slightly to ~ 0.085 before gradually falling off toward the specimen's far edge. The position of the strain peak makes it clear that the notch was on the left side of the camera's field of view.

Estimating and Reporting Strain Right at the ABS Notch, Right Before Failure

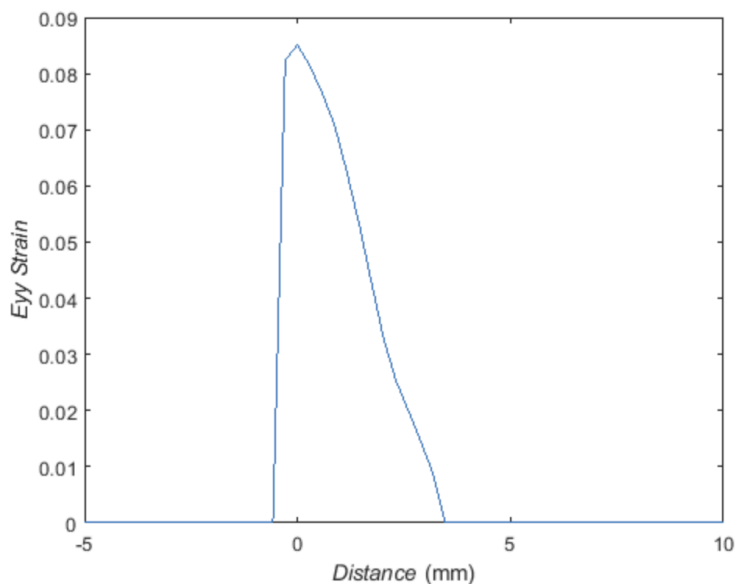


Figure 6. Plot of ϵ_{yy} along a horizontal transect immediately beside the notch in the notched TPU specimen at the final pre-failure frame. A sharp spike to roughly 0.45 strain pinpoints the notch location, after which the strain steadily tapers off toward the opposite edge of the sample. The position of this pronounced peak confirms that the notch was on the right side of the camera's view.

Discussion

a. Differences in the strain maps

- **Un-notched vs. Notched ABS:** The un-notched ABS sample spreads strain fairly evenly along its length, with a maximum $\epsilon = 0.0991$ just before failure. By contrast, the notched ABS concentrates almost all of its strain at the notch tip, forming a narrow plastic zone while the rest of the sample remains relatively undeformed. Although both reach the same peak strain (consistent with the material's failure limit), the notched geometry forces that strain into a much smaller region.
- **Notched ABS vs. Notched TPU:** The TPU sample endures far larger deformations at its notch (max $\epsilon \approx 5.464$) than ABS (max $\epsilon \approx 0.1029$) before breaking, reflecting TPU's higher ductility. Its strain-map hotspot is also broader, showing that the plastic zone extends farther from the notch. In ABS, the plastic zone is tighter and more localized.

b. Differences in the notch-adjacent strain profiles

- **ABS profile:** A sharp jump to $\approx 0.08 \epsilon$ appears right at the left-side notch, followed by a steep drop across the rest of the line. This matches the ABS map's narrow, high-contrast band at the notch.
- **TPU profile:** The TPU curve spikes to $\approx 0.45 \epsilon$ at the right-side notch and then tapers off slowly. This gentler slope corresponds to the wider, more gradual strain gradient seen in the TPU map.

In both cases, the direction of the spike (left for ABS, right for TPU) directly reflects the notch's position in the camera view, and the steepness of the slope reflects how sharply or broadly the material yields around the notch.